

Charlie Hayes

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I am looking for an internship for the summer of 2027. I am an organized and dependable candidate successful at managing multiple priorities with a positive attitude. Willingness to take on added responsibilities to meet team goals.

Experience

Xcel Energy

Summer 2026

Designed electrical power distribution systems for Xcel Energy's utility network. Scoped, investigated and designed distribution cases, including new customer service connections, grid reliability improvements and system safety enhancements. Managed multiple projects simultaneously to meet organizational workload demands. Collaborated with customers and field personnel to develop electrical distribution designs while meeting stringent project deadlines.

Automatic Systems Co.

Summer 2025

Worked in several areas of controls design automation for wastewater treatment plants including but not limited to: Switchgear and Power Distribution, Motor Control (Motor Control Centers, Variable Frequency Drives, Soft Starters, Motor Starters), Instrumentation and Controls, Networking and Communications and SCADA.

Vessco Inc.

Aug. 2025

Fabricated chemical skids for prolonged precise measurement and distribution of various chemicals for water treatment plants.

Education

IOWA STATE UNIVERSITY

Bachelor of Science in Electrical Engineering, Minor in Engineering Sales, GPA: 3.87

Additional Honors: Dean's List

Expected Graduation: Spring 2028

Skills

Engineering Software: SOLIDWORKS Electrical, GIS(Smallworld), AutoCAD, LTspice, Fusion 360

Laboratory: Oscilloscopes, Function Generators, Digital Multimeters, Soldering, PCB design

Programming: MATLAB, C, Python, Java, Arduino

Manufacturing & Business: SAP, Salesforce, Excel, BOM Management, Project Management

PROJECTS

- Automatic ball balancing robot: Balanced ball in 2d plane Mar. 2025
- HomeLab: Self-hosted personal WiFi and other IT infrastructure Jan. 2025
- Arduino builds: Speakers, Amps, Temperature and time sensor, Reaction time tester, Mini Radar built all from scratch myself, some on my own, others in Arduino club Nov. 2024
- FPV racing drone: designed, programmed and soldered Jan. 2023

Volunteering & Other Work

- Peer mentor (TA) for EE 1850 at Iowa State University
- Led youth football camps by facilitating and managing all camp activities for youth players
- Ambassador at Iowa State University Career fair